Automated Phishing Detection & Analysis System

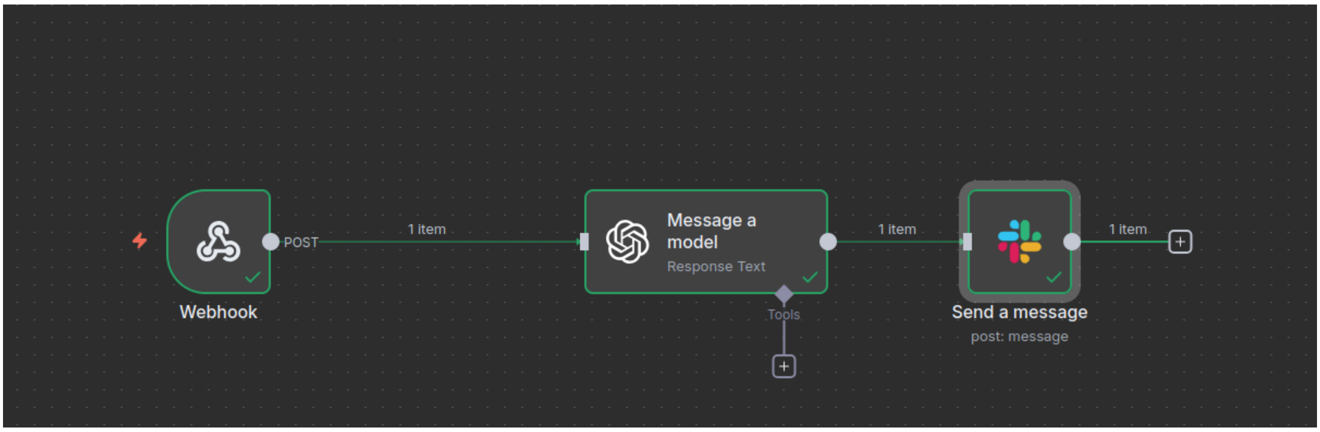
An intelligent automation workflow that integrates Splunk, n8n, and ChatGPT API to detect, analyze, and notify security teams about phishing attempts in real-time.

🎯 Overview

This project automates the detection and analysis of phishing emails using a three-stage pipeline:

1. **Detection**: Splunk monitors email traffic and identifies potential phishing attempts
2. **Analysis**: ChatGPT (GPT-4.1-MINI) performs deep analysis of suspicious emails
3. **Notification**: Slack alerts notify security teams with detailed analysis and recommendations

🏗️ Architecture

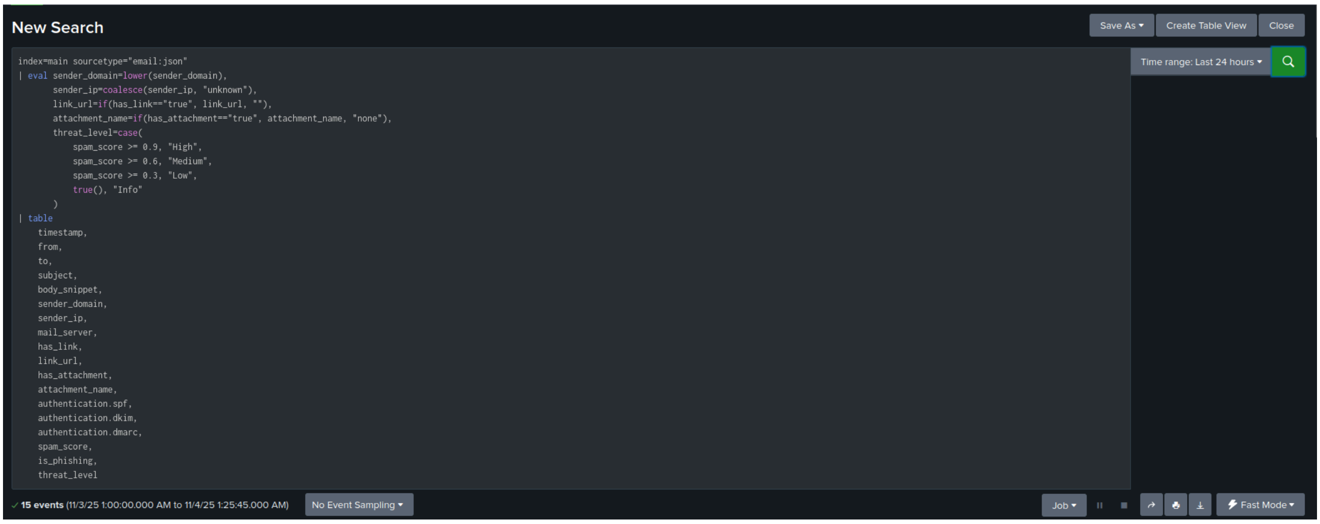


Workflow Components

1. **Splunk**: Email security monitoring and alerting
2. **n8n**: Workflow automation and orchestration
3. **ChatGPT API**: AI-powered phishing analysis
4. **Slack**: Real-time team notifications

⚙️ Configuration

Splunk Alert Configuration



Webhook Configuration:

- Set webhook URL to your n8n instance: `http://[n8n-host]:5678/webhook/phishing-alert`
- Method: POST
- Format: JSON

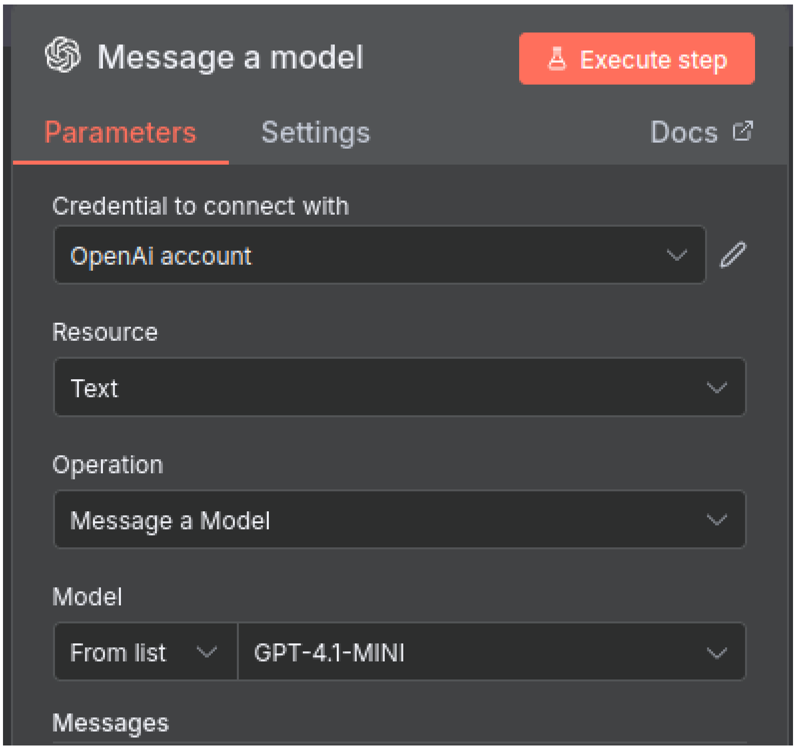
n8n Workflow Configuration

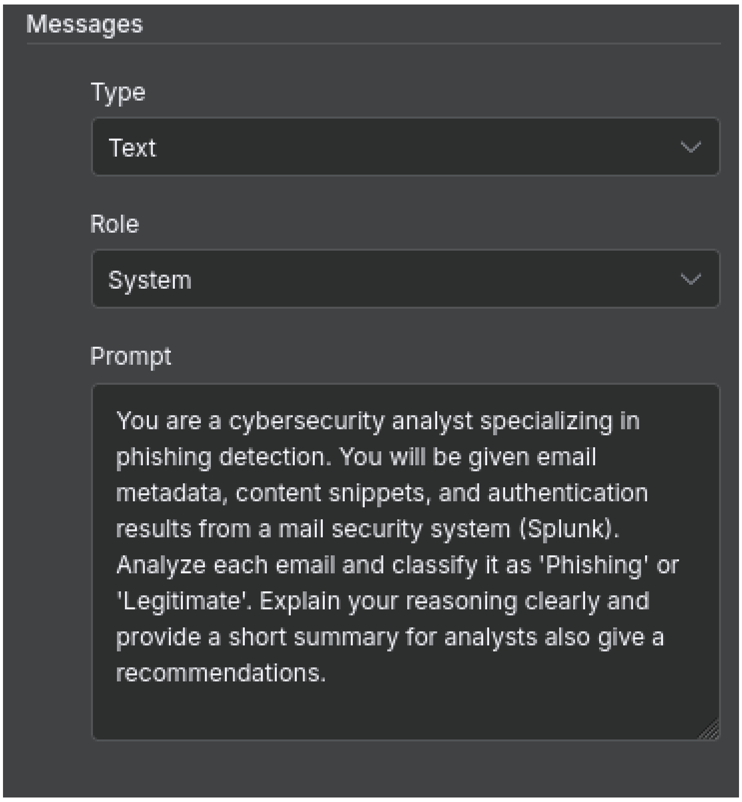
Node 1: Webhook Trigger

- **URL Path**: `/webhook/phishing-alert`
- **Method**: POST
- **Authentication**: None

Node 2: Message a Model (ChatGPT)

- **Credential**: OpenAI account
- **Resource**: Text
- **Operation**: Message a Model
- **Model**: GPT-4.1-MINI
- **System Prompt**:

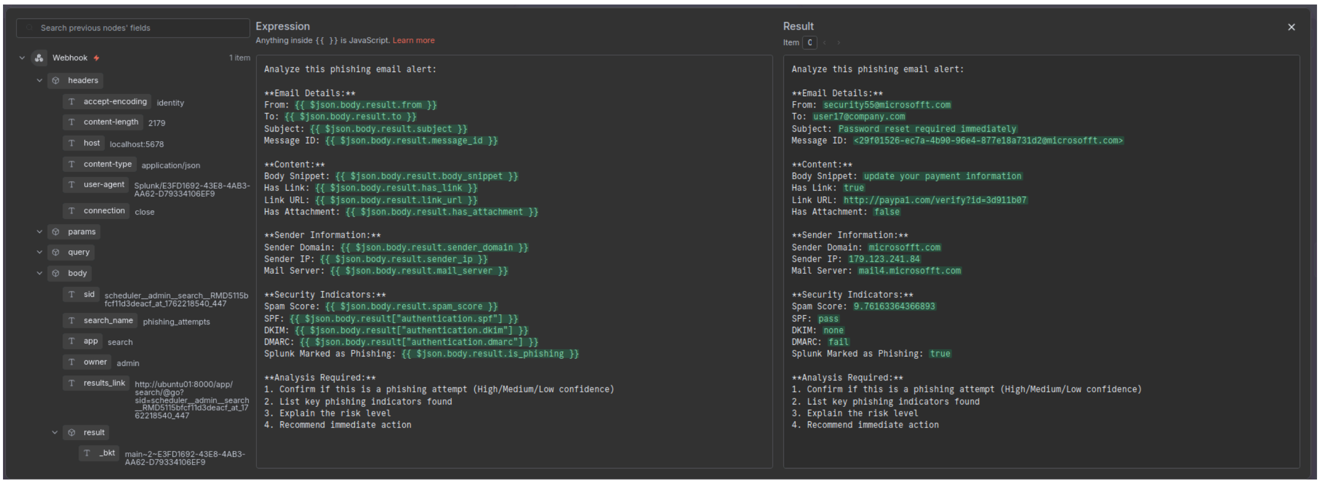




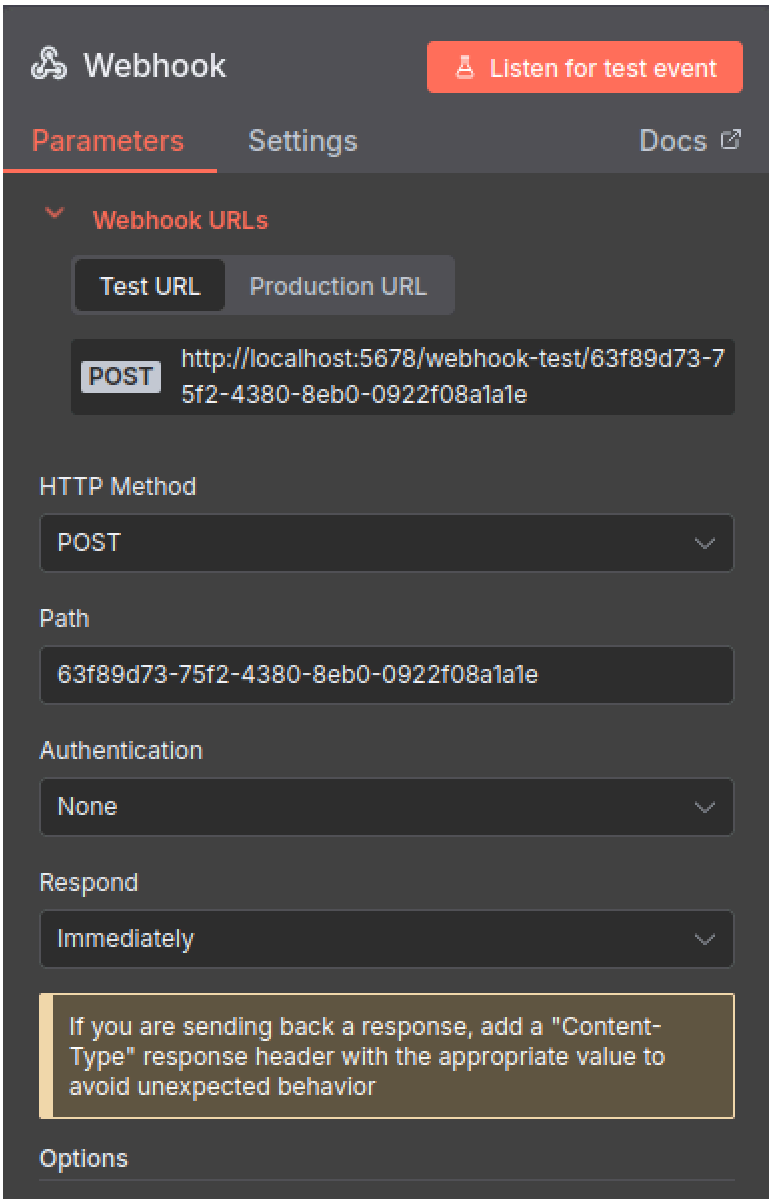
Node 3: Send Slack Message

- **Credential**: Slack account
- **Resource**: Message
- **Operation**: Send
- **Channel**: Select your security channel
- **Message Format**

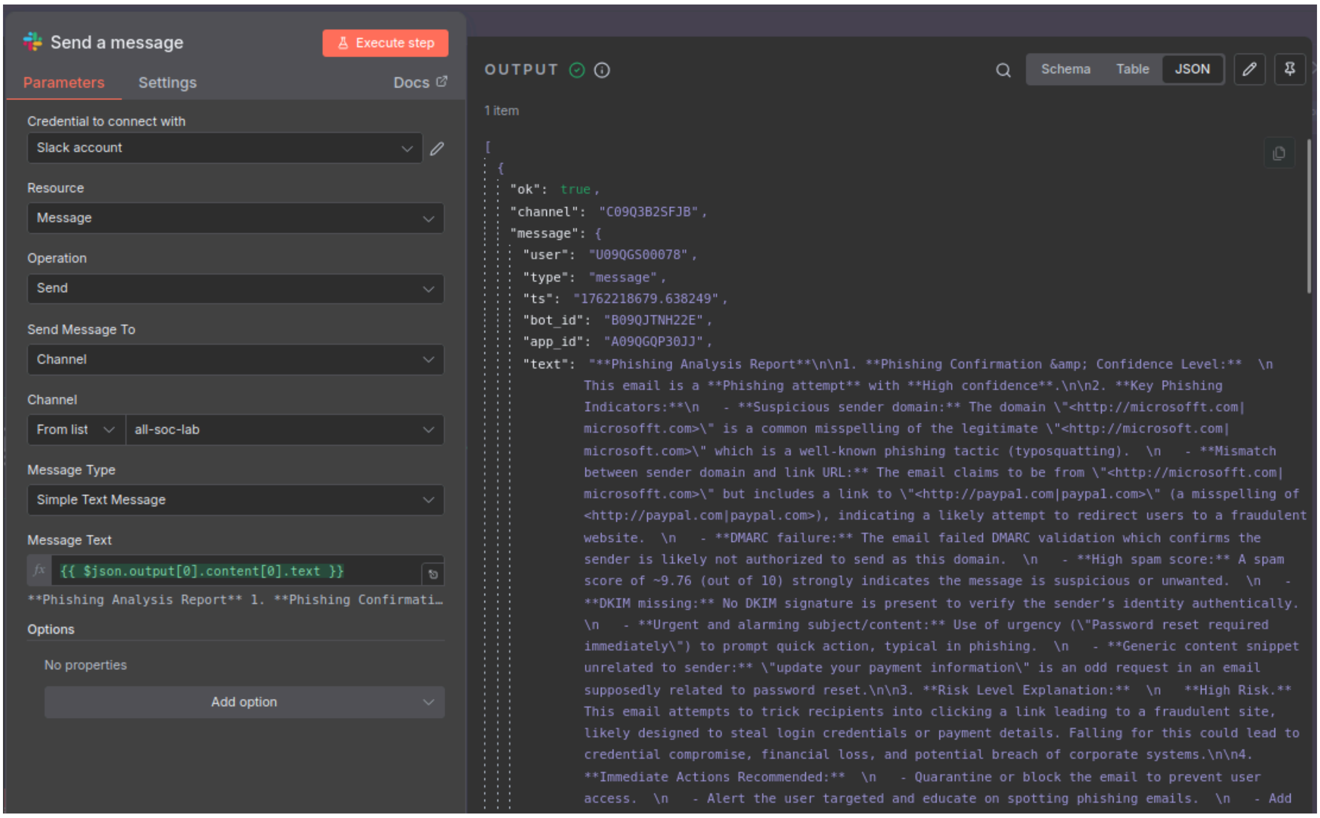
The following image illustrates the Data Extraction and Analysis



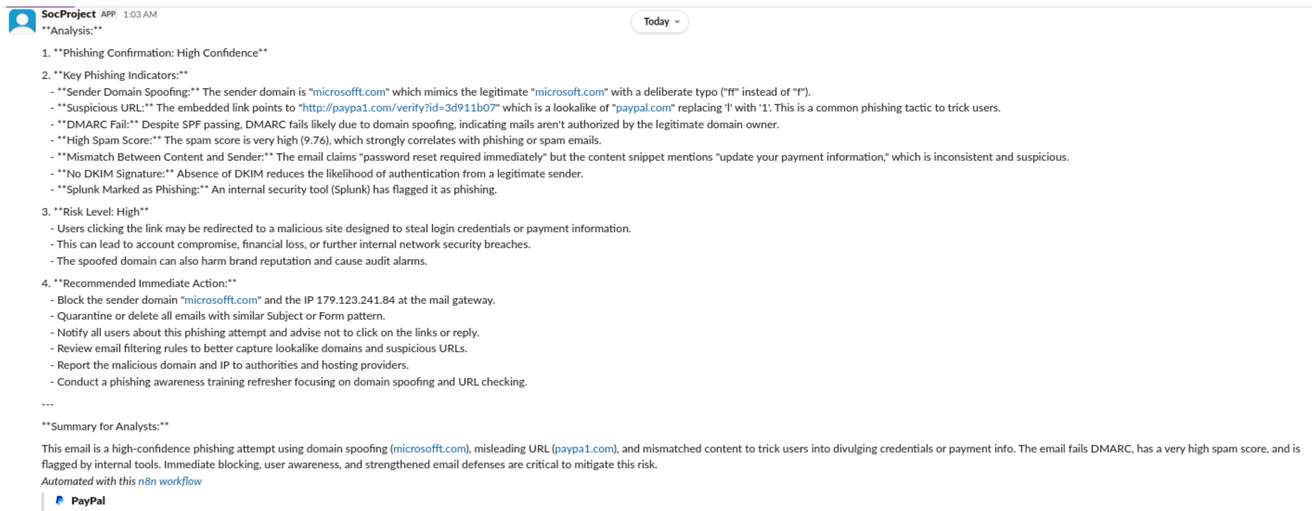
Webhook Configuration as shown in the above image



Testing with slack



Slack Notification Example



🔄 Workflow Details

Data Flow

1. **Splunk Detection Phase**:
   - Monitors incoming emails
   - Analyzes SPF, DKIM, DMARC authentication
   - Checks spam scores and suspicious patterns
   - Triggers webhook when phishing detected
2. **n8n Processing Phase**:
   - Receives webhook from Splunk
   - Extracts email metadata:
     - Sender domain and IP

- Subject line
    - Body snippet
    - Link URLs
    - Authentication results
    - Security indicators
3. **ChatGPT Analysis Phase**:
    - Analyzes email characteristics
    - Identifies phishing indicators:
        - Domain spoofing
        - Suspicious URLs
        - DMARC failures
        - High spam scores
        - Content mismatches
    - Provides confidence level assessment
    - Generates actionable recommendations
4. **Slack Notification Phase**:
    - Formats analysis results
    - Sends to designated security channel
    - Includes all relevant details for incident response

📈 Future Enhancements

- [ ] Add automated response actions (quarantine emails)
- [ ] Implement machine learning model for local analysis
- [ ] Create dashboard for phishing trends

🙏 Acknowledgments

- Splunk for robust security monitoring capabilities
- n8n for flexible workflow automation
- OpenAI for powerful AI analysis capabilities
- Slack for seamless team communication